



Cell Lysis for Metal Binding Proteins

Designing Life with AI

July 14, 2026

Description

This method allows us to recover binding proteins produced by our engineered bacteria, which are too large to be secreted and therefore require cell disruption for extraction.

Material

- Lysis buffer components
 - Sodium phosphate buffer, pH 8.5 (50 mM)
 - NaCl (300 mM)
 - Lysozyme (0.5 mg/ml final concentration)
 - PMSF (Phenylmethylsulphonyl fluoride, 1 mM final concentration — prepare fresh or from a 100 mM stock in isopropanol)
 - ddH₂O (to adjust final volume to 5 ml)
- Culture medium (LB or 2xTY)
- 70% ethanol
- Sonicator with MS37 probe

Steps

After inducing expression in a 10 ml culture for 24h in a 50 ml falcon tube, proceed with the following steps:

1. Centrifuge the tube(s) at 5'000 rpm, 4 °C for 10 min.
2. Prepare the lysis buffer following this recipe, made for a 5 mL volume for example:

Ingredients	Concentration	Volume for 5 mL solution
Sodium phosphate pH 8.5	50 mM	1 ml
NaCl	300 mM	1 ml
Lysozyme	0.5 mg/ml	50 µl
Phenylmethylsulphonyl fluoride (PMSF)	1 mM	250 µl
ddH ₂ O	—	2.7 ml

3. Pour out the supernatant.

The next steps are performed on ice.

4. Add 1 mL of lysis buffer to the cell pellet and resuspend the pellet.
5. If multiple culture tubes were prepared, transfer all the resuspended cells in one tube.
6. Put the 50 ml falcon tube in a big becker full of ice.
7. Sonificate the solution.
 - (a) Insert the previously disinfected MS37 probe (make sure it's tightly fixed) into the sample: center the probe in the tube without touching the bottom or sides.
 - (b) Start the sonicator and set the following parameter:
 - i. Pulse at 0.001.0S, 001.0S
 - ii. Amplitude at 40%
 - iii. Set for a 3 to 5 min programm

- (c) Sonicate the sample for a time corresponding to twice the volume of the lysis buffer used.
 - (d) Wait 2 minutes between each program before restarting.
8. After sonication, visually check the sample: a successful sonication often appears lighter/whiter than the unsonicated suspension.
 9. Centrifuge the tube at 5'000 rpm, 4 °C for 10 min.
 10. Keep the supernatant as all the binding proteins are floating in it whereas the cellular debris are in the pellet.